

Mathematical models to describe fixation disparity curves

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Abstract

Purpose: This study developed mathematical models to describe the four classical types of Fixation Disparity Curves (FDCs), addressing the limitations of existing generalised polynomial fits. These models provide an objective framework for analysing FDCs, with the goal of improving the diagnosis and management of binocular vision disorders.

Methods: The study was conducted in two phases. In the first phase, elementary mathematical functions were identified for each type of FDC: a polynomial function for Type I, an exponential function for Types II and III and a trigonometric function for Type IV. The function parameters were optimised using the least squares method in MATLAB. In the second phase, the models were validated using experimental data from 20 participants to ensure representation of all FDC types. The fixation disparity was measured using the Wesson card over a range of seven vergence demands.

Results: The proposed models accurately represented the shape and characteristics of each FDC, achieving 85% classification accuracy across all four types. Agreement between the model and subjective classification by optometrists was 75%. A statistically significant difference was observed in slope calculations between types using the developed model ($p=0.002$), but not with a general polynomial fit ($p=0.36$).

Conclusions: The newly developed models improve the precision and reliability of FDC analysis, thereby reducing subjective bias. These models have potential applications in both clinical and research settings for more accurate binocular vision assessments.

KEYWORDS

binocular vision disorders, fixation disparity curves, mathematical models, vergence

INTRODUCTION

Fixation disparity, a subtle misalignment of the visual axes during binocular fixation, is a critical component in understanding binocular vision and its coordination mechanisms.^{1,2} Epidemiological studies have shown that fixation disparity occurs in various forms, with distinct types differing in prevalence and clinical significance.^{1–3} Of the four classical types of fixation disparity curves (FDCs) described in Ogle's foundational work, Type I is the most common, occurring in approximately 60% of individuals.³ These

curves, also known as forced vergence curves, represent the stress on the vergence response within Panum's fusional area under varying visual demands, such as when prisms or lenses are used to alter vergence or accommodative loads.^{4–6} Type I is characterised by a symmetrically sigmoid shape between base-in (BI) and base-out (BO) demands and reflects a normal response with high vergence adaptation in both directions. Type II, observed in approximately 25% of cases, shows an asymmetric sigmoid curve with reduced adaptation to BO demands and a normal BI response, whereas Type III (approximately 10%) shows the

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opposite pattern, reduced adaptation on BI and a normal response to BO. Both Types II and III are typically associated with binocular dysfunction and related visual symptoms.⁷ Type IV, the least common type (~5%), exhibits a nearly flat curve to either BI or BO demands.

The key features of these curves include their slope, shape and centre of symmetry. These characteristics provide valuable information about an individual's binocular visual system, including its adaptability and stability during stress. For example, the slope of the FDC reflects the rate of change in fixation disparity as vergence demand varies, serving as an indicator of the vergence system's adaptability.^{4,8} Curves with steep slopes are found more frequently in symptomatic patients.^{1,9} At the physiological level, fixation disparity arises from the complex interplay of sensory and motor components in binocular vision.^{10–12} These processes are shaped by vergence adaptation, accommodative responses and the limits of Panum's fusional area.^{7,13} As noted previously, the shape of the FDC reflects its relationship with visual symptoms and adaptive processes. The centre of symmetry, which represents the flattest central region of the FDC, corresponds to the point at which vergence adaptation responds most efficiently to changes in fusional vergence.

Clinically, FDCs have been used to diagnose and manage binocular vision anomalies such as convergence insufficiency, which can manifest as symptoms such as headaches, blurred vision and asthenopia,^{5,7,14} for reviews, see Carter¹⁵ and Fogt.¹⁶ It is important to note that in investigations using FDCs, the identification of FDC type in previous research has primarily been conducted subjectively.^{17–20} Nonetheless, some have focused primarily on fitting general functional forms to experimental data. A number of models used in research approximate FDCs using orthogonal polynomial functions to capture the basic shape of the curve and calculate the slope, which is usually the dependent variable of the study. Nevertheless, the selection of polynomial order varies across studies,^{4,21–23} as do the methods used for data analysis.

In this context, some studies have employed the Best PEST (Parameter Estimation by Sequential Testing) method, while others considered Probit analysis.^{4,8,9,21,22,24–26} Best PEST is particularly effective in determining threshold values in psychophysical research by adaptively adjusting the test parameters based on the participant's responses. However, the Best PEST method tends to exhibit a negative bias, which decreases as the number of trials increases. Consequently, this method requires more measurements to provide accurate threshold estimates compared with other approaches.²⁷

On the other hand, Probit analysis, a type of regression used to model binary or ordinal outcomes, is also often employed to understand the relationship between stimulus intensity and the probability of a particular response.²⁸ In the context of FDCs, Probit analysis helps to quantify vergence adaptation and fixation disparity in relation to different vergence demands. However, this

Key points

- This study developed type-specific mathematical models for fixation disparity curves using polynomial, exponential and trigonometric functions.
- The study validated the models with experimental data, achieving 85% accuracy.
- It provides enhanced objectivity in the analysis of fixation disparity curves, providing substantial potential for both clinical and research uses.

analysis is not well suited for certain datasets and may not be applicable in all cases.^{20,21}

Therefore, it seems that fitting FDCs could be improved by employing specific theoretical models. Each type of FDC has distinct mathematical functions; thus, a more robust model is required rather than using the same polynomial fitting for all shapes (types). The variability in the shapes of FDCs, such as the symmetric sigmoid curve in Type I or the flat segments observed in Types II and III, indicates that a single polynomial fitting may not accurately capture the nuanced behaviours of each type. Moreover, developing a theoretical and mathematical model for all types of FDCs could avoid subjective interpretation and instead rely on scientific rigour. This study aims to overcome the limitations of existing models by developing a comprehensive mathematical framework that assigns an appropriate elementary function to each type of FDC. By incorporating these mathematical models, the study seeks to improve the objectivity and reliability of FDC analysis. To ensure its practical relevance, the developed models will be validated against experimental data to demonstrate the capacity of the proposed model to represent real fixation disparity behaviour and to compare it with established subjective classification criteria.

METHODS

This investigation was based on two primary studies. The first aimed to develop mathematical models for each FDC shape. The four distinct types of FDCs were classified using elementary functions, considering the main physiological components such as the centre of symmetry. The optimal elementary function for each FDC type was determined by minimising the error using the least squares method.²⁹ All calculations were performed using MATLAB ([mathworks.com](https://www.mathworks.com)).

The second study evaluated the robustness of the mathematical model by comparing its predictions with real-life data. Participants were recruited for this analysis, all university students, without strabismus or amblyopia, ensuring a visual acuity at far and near with their habitual refractive

correction of at least 0.00 logMAR with an Early Treatment of Diabetic Retinopathy (ETDRS) visual chart (bernell.com) and ≤ 70 arc seconds of stereoacuity (Random Dot Test, stereooptical.com). To ensure the analysis of different shapes of FDCs, rather than just Type I (normal response without visual symptoms), the optometric data collected did not include near-point of convergence or decompensated heterophoria, as these have been associated with other FDC shapes.^{1,5,7} The analysis aimed to identify at least one FDC for each type (I, II, III and IV) during the recruiting phase.

Participants who satisfied the inclusion criteria had their FDC determined. The Wesson Fixation Disparity Card (bernell.com) was used to measure subjective FDC using $-15, -10, -5, 0, 5, 10$ and 15Δ of vergence demand (negative as base-in prism, positive as base-out prism) at 40 cm. At this distance, each 1 mm of fixation disparity corresponds with 0.25 prism dioptres (Δ), or 8.6 arcmin ($'$) of measurement, with a maximum level of 1Δ , or $34.40'$. Participants were not allowed to hold the Wesson Fixation Disparity Card during the measures, were tested in primary gaze and used a chin-rest to maintain the test distance. The habitual correction (either glasses or contact lenses) was worn. After every change of prism a 5 s rest was allowed, with prism

values starting from 0, followed by $-5/5, -10/10$ and $-15/15 \Delta$. All verbal responses were recorded in a spreadsheet. To ensure data integrity for the purposes of this study, participants were included who demonstrated a fixation disparity response for all tested vergence demands. All participants were fully informed about the nature of the study and provided written consent prior to participation. The study protocol was reviewed and approved by the Institutional Review Board of the Universitat Politècnica de Catalunya.

RESULTS

Study 1: Development of the mathematical models

Figure 1 shows Types I, II, III and IV FDCs. After considering some elementary mathematical functions, a third-degree polynomial was associated with the shape of Type I, two exponential functions for Types II and III and a trigonometric function for Type IV.

Preliminary testing of unconstrained fits revealed that, in several cases, exponential or trigonometric functions could degenerate into near-linear curves, which do not

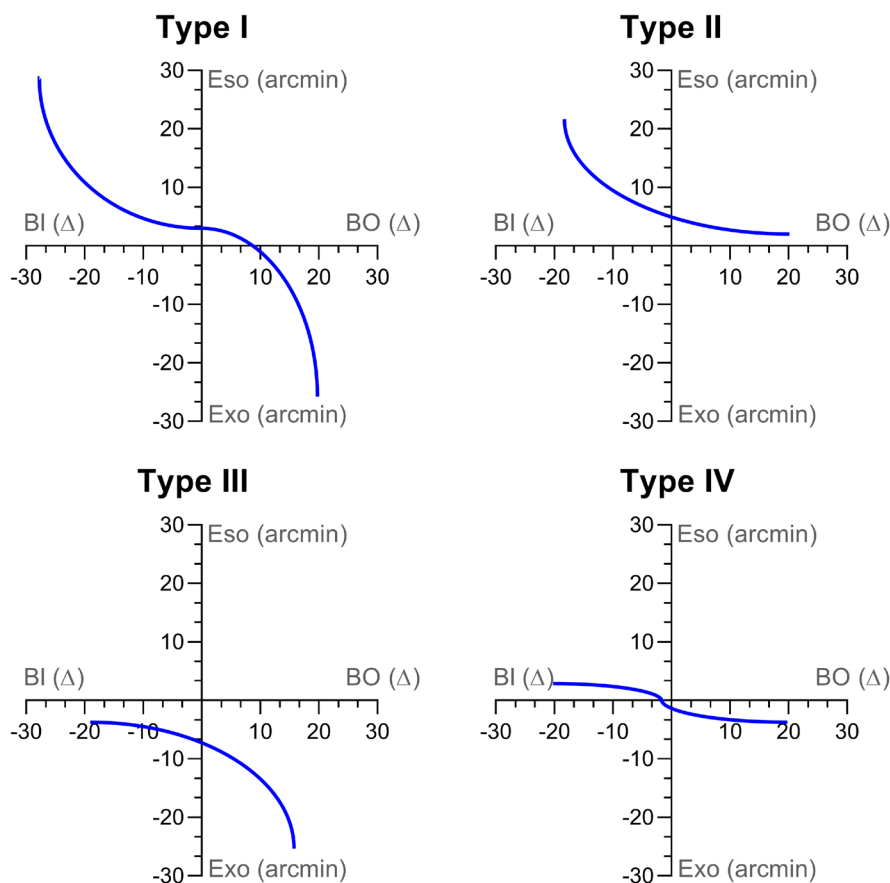


FIGURE 1 The four Fixation Disparity Curves (FDCs) described as Types I, II, III and IV. These plots are based on the classical classification proposed by Ogle.³ The X-axis represents vergence demand in prism dioptres (Δ), with BI representing Base In, BO representing Base Out and the Y-axis representing fixation disparity in minutes of arc (arcmin). At 40 cm, 1 mm corresponds to 8.6 arcmin, which is equivalent to 0.25Δ .

reflect the physiological behaviour described for the different FDC types. Such cases introduce ambiguity into the classification process and reduce clinical interpretability. Hence, a boundary on established physiological ranges was added for fixation disparity and the centre of symmetry,^{1-3,5,6,19} ensuring that the modelled curves remain consistent with all four FDC shapes.

In fact, taking into account the centre of symmetry for Types I and IV, as well as upper and lower bounds for Types II, III and IV, the following elementary functions were established:

- Type I is described by

$$f_1(x) = a_0 + a_1 \cdot (x - a_2)^3,$$

- Type II follows the exponential function

$$f_2(x) = a_0 + a_1 \cdot e^{-a_2 \cdot (x - a_3)},$$

- Type III with the exponential function

$$f_3(x) = a_0 - a_1 \cdot e^{a_2 \cdot (x - a_3)},$$

- Type IV is estimated by the trigonometric function

$$f_4(x) = a_0 - a_1 \cdot \arctan(a_2 \cdot (x - a_3)),$$

where a_0 , a_1 , a_2 and a_3 are model parameters: a_0 denotes vertical displacement; a_1 represents vertical expansion and contraction; a_2 (for Type I) and a_3 (for the other types) correspond to horizontal displacement and a_2 (for Types II, III and IV) accounts for a certain degree of horizontal expansion and contraction, as appropriate. According to the clinical data presented in the literature, the centre of symmetry for Types I and IV typically falls within the range of -10 to $+10$ arcmin.^{1,3,5} Similarly, for Type II, the minimum value is generally lower-bounded by -10 arcmin, while for Type III, the maximum value is upper-bounded by $+10$ arcmin. Based on these assumptions, Type I considers $-10 \leq a_0, a_2 \leq 10$, and the centre of symmetry is established in the square formed by the range of x and $f_1(x)$ values between -10 and 10 . In the same vein, Type IV accounts for $-10 \leq a_0, a_3 \leq 10$. On the other hand, Type II establishes $f_2(x) \geq -10$ and its fixation disparity is less than or equal to -10 . Eventually, for Type III, $f_3(x) \leq 10$ for fixation disparities smaller than or equal to 10 . For Type IV, these values are between -10 and 10 , that is, $-10 \leq f_4(x) \leq 10$. Finally, for Types II, III and IV, it was assumed that $a_1, a_2 \geq 0.1$ to ensure close alignment with the corresponding exponential or trigonometric function rather than a linear or constant function.

Now, given experimental data $(x_1, y_1), (x_2, y_2), \dots, (x_k, y_k)$, one can compute the coefficients a_0, a_1, a_2, a_3 according to the least square method.²⁹ In other words, the coefficients were found to minimise the root mean square error (RMSE) given by the following formula:

$$\sqrt{\frac{\sum_{j=1}^k (f_i(x_j) - y_j)^2}{k}} \quad (1)$$

for $i = 1, 2, 3, 4$, assuming the previously defined restrictions for each function. To solve these systems, the *fmincon* package integrated in MATLAB was used (interested readers may request further details from the authors).

As detailed in the introduction, the slope calculation for FDC is an important parameter used in research to evaluate the change in fixation disparity,^{6,19} and is considered to be the best indicator of visual symptoms.⁷ Some authors have calculated the slope based on the changes of fixation disparity between 3Δ BI and 3Δ BO,^{2,19,30} using Formula 2. However, others calculated the slope using a straight regression line.^{9,26} Formally, in mathematics, it is important to note that this value represents the variation of the fixation value with respect to the vergence demand, rather than the slope, as it is defined at a specific point. Nevertheless, in this work, we continue to use the term 'slope' to refer to the local variation in fixation as a function of the vergence demand. Although this is not the slope of the fitted regression line, the term is retained for consistency with prior literature.^{9,26} It is important to clarify that in a mathematical approach, 'slope' describes the instantaneous rate of change (i.e., derivative) of the fixation disparity curve at a given point, rather than the global slope of a linear fit. Thus, Formula 2 effectively provides the slope of the line connecting points FD^3 BI and FD^3 BO.

$$\text{Slope} = \frac{|FD^3 \text{ BI} - FD^3 \text{ BO}|}{6} \quad (2)$$

where FD is the fixation disparity in arcmin. From this model, the slopes are determined by the following formula, $\text{Slope}_i = \frac{|f_i(3) - f_i(-3)|}{6}$ where $i = 1, 2, 3, 4$ correspond to Types I, II, III and IV, respectively.

Next, this model was tested using a theoretical example. Figure 2 shows the data of random values where x is the vergence demand and y is the fixation disparity, with x/y being $-15.0/25.8, -10.0/17.2, -5.0/12.9, 0.0/0.0, 5.0/-4.3, 10.0/-4.3$ and $15.0/-4.3$, (positive and negative values of x indicate BO and BI, respectively). Figure 2 illustrates the four corresponding types of functions obtained from the proposed model.

The results according to the model are given by $f_1(x) = -1.76 - 0.002 \cdot (x - 10)^3$, $f_2(x) = -6.05 + 0.86 \cdot e^{-0.1 \cdot (x - 21.94)}$, $f_3(x) = 11.01 - 1.82 \cdot e^{0.1 \cdot (x + 9.05)}$ and $f_4(x) = 2.84 - 4.58 \cdot \arctan(10 \cdot (x + 0.07))$, where the associated errors given by Formula 1 are 3.86, 3.02, 8.20 and 6.66 for Types I, II, III and IV, respectively. Thus, based on the simulation model, these results align with a Type II fixation disparity. Moreover, for the best-fit approximation, that is, Type II, the corresponding slope is $\text{Slope}_2 = \frac{|f_2(3) - f_2(-3)|}{6} = 0.785$.

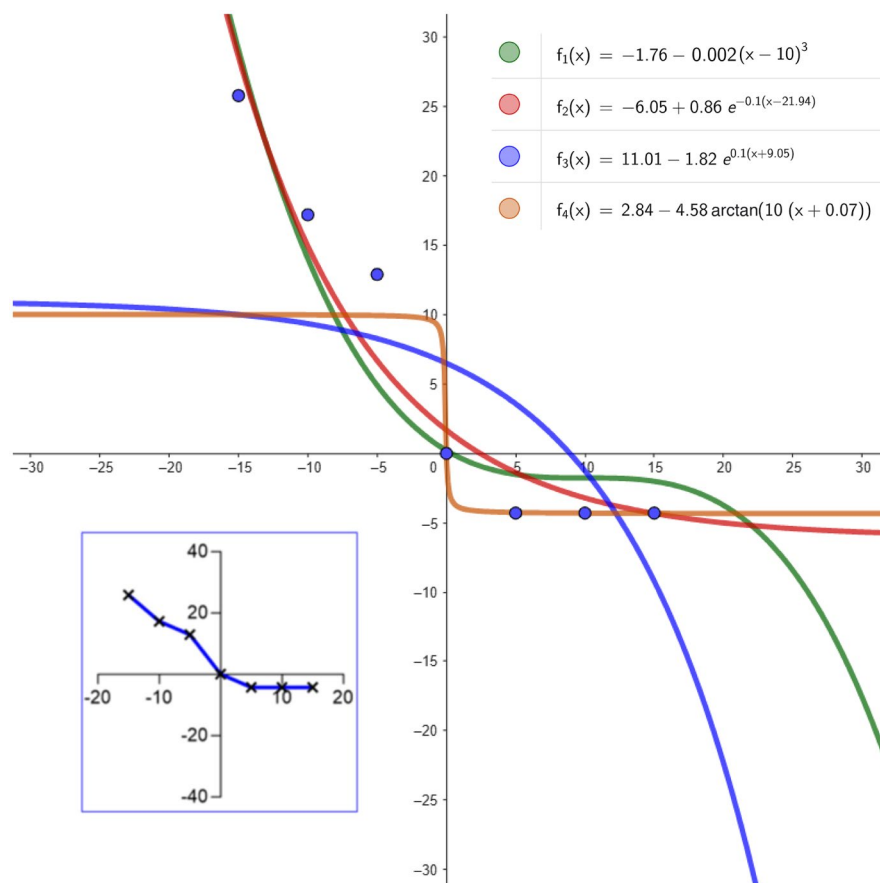


FIGURE 2 The four fixation disparity curves (FDCs) simulated in the example are Type I ' $f_1(x)$ ', II ' $f_2(x)$ ', III ' $f_3(x)$ ' and IV ' $f_4(x)$ '. Blue dots correspond to a random fixation disparity response using $-15, -10, -5, 0, 5, 10$ and 15 prism dioptres (Δ), with negative and positive indicating base-in prism and base-out prism, respectively. The inset plot shows the FDC response for the example.

Study 2: Validation for the models

The second phase of the study focused on validating the proposed mathematical models with empirical data, as model accuracy and classification reliability are essential for clinical and research applications. To validate the model with real data, 35 university students were screened for eligibility and 20 were included in the study. Fifteen students were excluded because they did not complete the full range of vergence demands during the fixation disparity test, either due to experiencing diplopia or suppression. All fixation disparity curves from the 20 participants are shown in Figure 3. Each participant was denoted as P_i , with $i = 1, \dots, 20$. The mean age and standard deviation ($\pm$) of the participants was 26.75 ± 4.50 years (range, 23–39 years), with a female/male ratio of 1:1.

First, to test the model, three experienced optometrists subjectively identified the types of FDCs within the 20 participants: 13 (65%), 1 (5%), 2 (10%) and 4 (20%) had Types I, II, III and IV, respectively. In addition, three professors in mathematics were selected to classify the 20 FDCs based on Figure 1. They were told to classify it based on their knowledge of algebraic functions. All three mathematicians

agreed to classify the 20 FDCs. Interestingly, there was no agreement between the subjective classification by optometrists and mathematicians in five cases (25%), specifically P_2, P_7, P_{12}, P_{14} and P_{17} . The inter-rater reliability between the subjective classifications from the optometrists was computed using Cronbach's alpha and yielded $\alpha = 0.44, p = 0.06$ and $\alpha = 1.00, p < 0.001$ from the mathematicians. The subjective agreement classifications for each participant can be found in Table S1.

Second, the developed model was compared by applying the least squares method, as described previously, to fit each elementary function across all types of FDCs, thereby determining the corresponding coefficients. Table 1 summarises the errors (using Formula 1) associated with each approximation described above according to the four types for all 20 participants. The best-fit approximation for each participant is highlighted in bold. In all cases, the smaller the error, the better the fit. In addition, Table 1 shows colour coding to represent the percentage of relative error for the best approximation ($Er_{\min}$) for each participant, $\%ErT_i = [(ErT_i - Er_{\min})/Er_{\min}] \times 100$, for $i = 1, \dots, 20$. Note that ErT_1 = Error Type I, ErT_2 = Error Type II, ErT_3 = Error Type III, ErT_4 = Error Type IV. P_i denotes the i -th Participant, where $i = 1, \dots, 20$. Errors are subdivided into $<10\%$, between 10

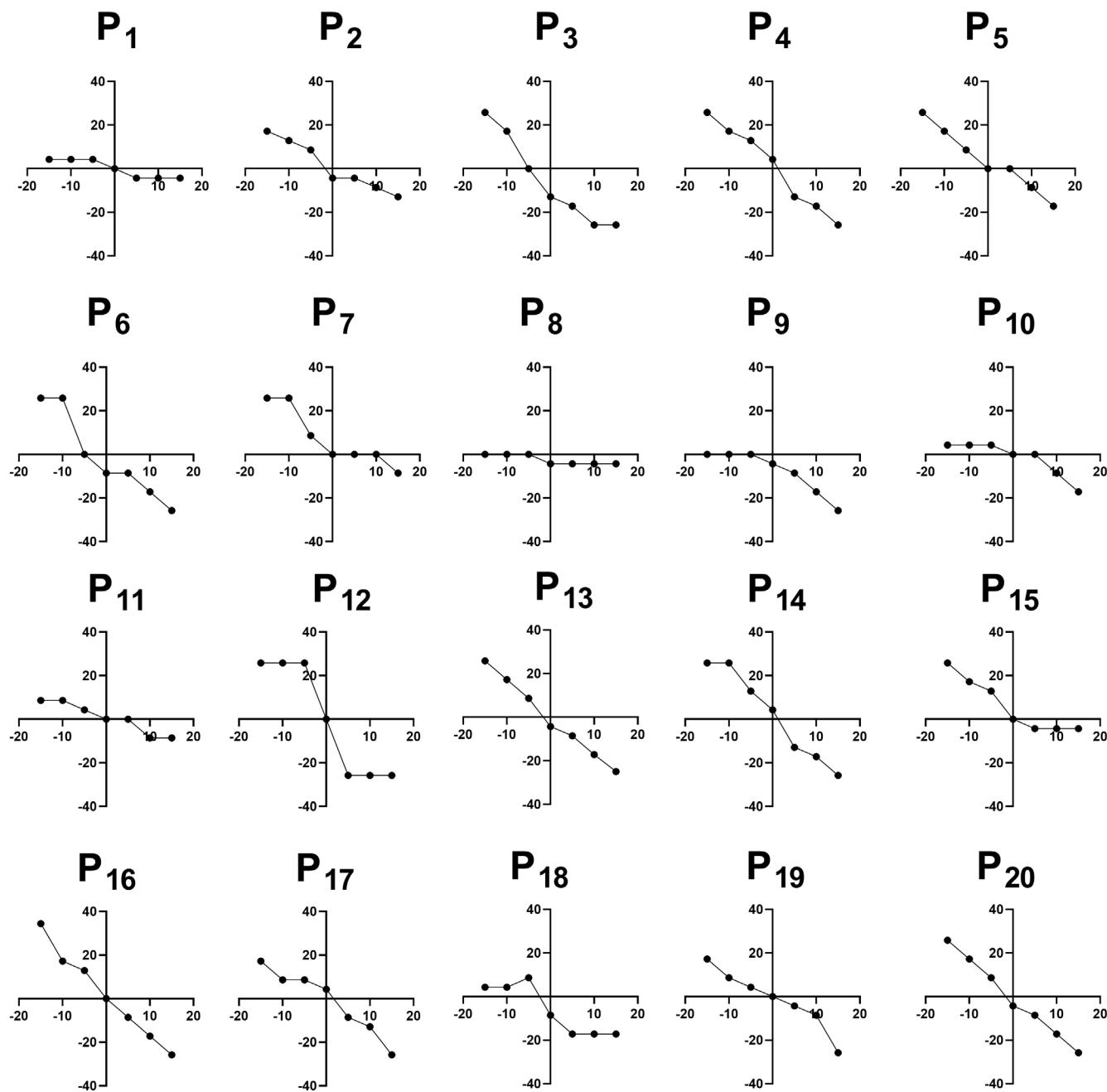


FIGURE 3 Fixation disparity curves from the 20 participants (P) included in the study. The X-axis represents the vergence demand in prism dioptres (negative for Base In, positive for Base Out). The Y-axis represents the fixation disparity in arcminutes (negative for exophoria, positive for esophoria).

and 50%, between 50% and 90% and >90%. It was considered that the model correctly approximates the type of FDC for a value <10%.

It may be concluded that while in some cases the method provides clear criteria, it can be confusing in others (e.g., P_4 , P_{12} and P_{14}). Although all cases fell below 10% relative error, which was considered the minimum error, as shown in Table 1. In general, the model simulates each type of FDC in 17/20 cases (85%), leaving 3/20 cases (15%) without successful modelling. For the new model, P_4 aligns with Type I, P_{12} with Type IV and P_{14} with Type II (see Table 1 for errors).

The agreement, when compared with the subjective classification was 15/20 cases (75%), leaving five cases without agreement, specifically, P_2 , P_7 , P_{12} , P_{14} and P_{17} . Readers can evaluate the subjective criteria in Table S1. Indeed, the subjective criteria given by the experienced optometrist had an 100% agreement with Types II, III and IV. However, Type I had a lower agreement of 8/13 (61.5%).

Some controversial cases are discussed in greater detail below. Table 2 shows the experimental data of participants 12 (P_{12}) and 14 (P_{14}), where 'x' is the vergence demand and 'y' is the fixation disparity.

TABLE 1 Errors associated with the analysis for Types I, II, III and IV fixation disparity curves applied to the 20 participants included in the study.

	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉	P ₁₀
ErT ₁	2.69	4.86	7.89	8.10	4.03	8.05	5.57	1.30	1.63	1.16
ErT ₂	2.25	3.91	10.20	8.74	4.64	8.82	4.62	1.18	7.23	5.57
ErT ₃	2.25	5.44	11.26	8.37	7.03	10.42	9.08	1.49	1.06	1.07
ErT ₄	0.01	3.63	11.11	9.42	7.47	10.73	8.87	0.02	6.58	2.94
	P ₁₁	P ₁₂	P ₁₃	P ₁₄	P ₁₅	P ₁₆	P ₁₇	P ₁₈	P ₁₉	P ₂₀
ErT ₁	3.32	16.13	6.50	9.69	3.86	6.73	4.70	7.80	2.24	6.47
ErT ₂	3.46	14.88	7.21	9.61	3.02	7.51	7.78	7.36	6.61	7.45
ErT ₃	2.72	14.88	7.96	10.08	8.20	10.53	3.73	6.56	3.40	7.89
ErT ₄	1.56	14.66	9.12	10.82	6.66	11.71	6.69	4.92	6.57	9.31

Note: ErT₁ = Error Type I, ErT₂ = Error Type II, ErT₃ = Error Type III and ErT₄ = Error Type IV. P_i denotes the *i*-th Participant, where *i* = 1, ..., 20. In bold is shown the minimum error, denoted as Er_{min}. Colours associated with the percentage of relative errors in the analysis for Types I (%ErT₁), II (%ErT₂), III (%ErT₃) and IV (%ErT₄). Dark green indicates %ErT_i < 10%, light green indicates %ErT_i between 10% and 50%, orange between 50% and 90% and red indicates error percentages > 90%.

For P₁₂, the associated errors were 16.13, 14.88, 14.88 and 14.66 for Types I, II, III and IV, respectively, and the percentage of relative errors was <10% for Types II, III and IV (see Table 1). Thus, based on the analysis, these results fail to align properly with any of the approximations. However, the least inaccurate approximation corresponded to Type IV, exhibiting a slightly lower error, although the subjective classification from the optometrist was Type I. Figure 4 (Panel a) presents the function plots for this participant.

Similarly, Table 2 shows the experimental data for P₁₄. The corresponding functions according to the model approximations are shown in Figure 4. The associated errors were 9.69, 9.61, 10.08 and 10.82 for Types I, II, III and IV, respectively (see Table 1). It can be seen that these results do not fit well with the approximations between all types of FDC (see Table 1). For instance, the model classifies P₁₄ as Type II, because it shows a marginally lower error, but was categorised subjectively as Type I by the optometrist. Note that these results are based on the constraint of ±10 arcmin applied to fixation disparity. While other classifications may lead to different outcomes under alternative constraints, these restrictions were chosen specifically as described in the development of the mathematical model (Study 1).

Since the slope is fundamental to analyse the properties of FDC, the slopes were calculated to examine the differences between the developed model and those obtained using a general third-degree polynomial function. The slope from a general best-fit using a third-degree polynomial $y(x) = b_0 + b_1x + b_2x^2 + b_3x^3$, given by Formula 2, is $\text{Slope} = \frac{|y(3) - y(-3)|}{6} = |b_1 + 18 \cdot b_3|$.

The slopes calculated from the model applied to each type are given by $\text{Slope}_1 = |3a_1a_2^2 + 9a_1|$, $\text{Slope}_2 = \left| \frac{1}{3}a_1 e^{a_2a_3} \sinh(3a_2) \right|$, $\text{Slope}_3 = \left| \frac{1}{3}a_1 e^{-a_2a_3} \sinh(3a_2) \right|$ and $\text{Slope}_4 = \left| \frac{1}{6}a_1 \arctan \left(\frac{6a_2}{1 + a_2^2(a_3^2 - 9)} \right) \right|$, respectively. All the descriptive results can be found in Table S2.

TABLE 2 Experimental data of participants 12 (P₁₂) and 14 (P₁₄).

	P ₁₂	P ₁₄
x	y	y
-15.00	25.80	25.80
-10.00	25.80	25.80
-5.00	25.80	12.90
0.00	0.00	4.30
5.00	-25.80	-12.90
10.00	-25.80	-17.20
15.00	-25.80	-25.80

Note: 'x' values indicate the vergence demand in prism dioptres (Δ), with negative and positive values being base-in and base-out, respectively. The 'y' values are fixation disparity in arcmin ('), with negative and positive reflecting exo and eso fixation disparity, respectively.

In research, the slope is a key variable commonly used to compare groups or investigate relationships with clinical outcomes.^{4,21,22,23} Therefore, assessing whether there are differences in slope estimates between models is highly relevant. Moreover, based on the characteristics of the different types of fixation disparity curves, it is expected that slope values should differ across curve types; for example, Type IV curves are typically steeper than Type I. In the present study, the slope calculation was compared between the model and a general polynomial fit. The resulting slopes followed a non-parametric distribution: Shapiro–Wilk test, $p = 0.04$ and $p < 0.001$, for the general and the current model, respectively. The median for the general calculation was 1.28, interquartile range (IQR) (0.81, 1.83), and the present model was 0.66, IQR (0.08, 1.02). The Wilcoxon test showed significant differences between them ($p = 0.01$). In addition, a one-way Kruskal–Wallis test was performed to identify any FDC type interaction as a factor between slopes. There was no significant difference between the slopes and type of FDC with the general polynomial approximation, $p = 0.36$, but a significant difference was observed between the four

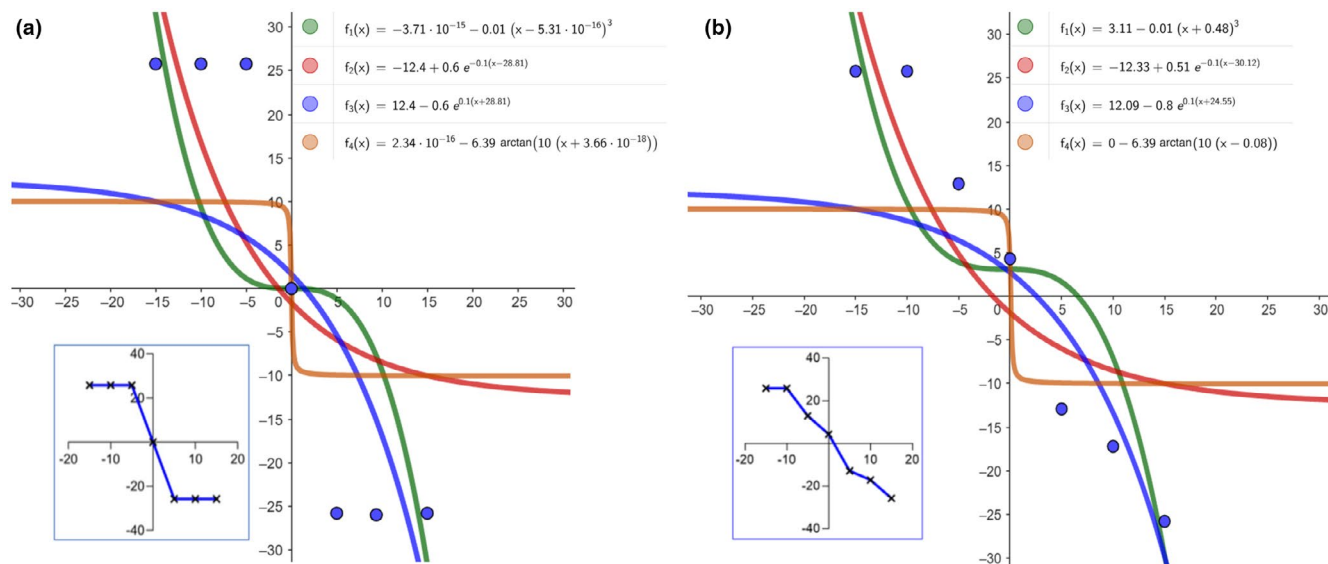


FIGURE 4 Panel a: The four fixation disparity curves (FDCs) associated for participant 12 (P_{12}) for Type I ' $f_1(x)$ ', II ' $f_2(x)$ ', III ' $f_3(x)$ ' and IV ' $f_4(x)$ '. Panel b: The four FDCs associated with participant 14 (P_{14}) for Type I ' $f_1(x)$ ', II ' $f_2(x)$ ', III ' $f_3(x)$ ' and IV ' $f_4(x)$ '. The blue dots correspond to the data from the Wesson disparity card. The individual FDCs from P_{12} and P_{14} are shown in the inset plots.

types of FDCs with the current model ($p=0.002$). Pairwise comparison with Bonferroni correction showed significant differences between Types I and IV ($p=0.007$), but not between the other types of FDCs ($p>0.10$ for all pairwise comparisons). Type IV slopes had higher values than others with the current model, but not using the general polynomial function. Descriptive data for the analysis for each Type of FDC and slope can be found in [Tables S2](#) and [S3](#).

DISCUSSION

This study aimed to develop a comprehensive mathematical model to characterise the four types of FDCs and to enhance the robustness of research and clinical analysis. To achieve this, a theoretical study was conducted to classify the four distinct types using elementary functions, considering both their shape and centre of symmetry. A second study validated the model using experimental data from 20 participants with different FDC shapes, and applied the developed model to classify all FDCs. Additionally, the subjective classification criteria of three optometrists and three mathematicians were compared with the model classifications.

It was determined that among the three optometrists, there was a discrepancy in the classification of FDC types, particularly for Type I. All optometrists were clinically and theoretically familiar with the concept of fixation disparity and FDCs. Interestingly, all the mathematicians reached subjective agreement among themselves for all 20 FDCs and showed 100% agreement with the model, whereas the optometrists agreed with the model in only 15 of the 20 cases (75%). Overall, the model accurately simulated each type of FDC in 17 out of 20 cases (85%), leaving some with

less clear classification criteria. This highlights the need for an objective model to classify FDCs.

Specifically, the shape classification for participants P_2 , P_7 , P_{12} , P_{14} and P_{17} did not align fully with the developed model or the subjective classification by the mathematicians. On the one hand, in cases P_2 , P_7 and P_{12} , it was confirmed that the model was consistent with the classical classification given by Ogle (see [Figure 1](#)). On the other hand, from a mathematical perspective, the classifications of P_{14} and P_{17} are clearly correct according to the developed model. Even within the model, some classifications exhibited minimum errors. For instance, for P_{14} , the error was 9.61, that is, <10%. Despite this, the model provided three classification options for P_{14} , namely Types I, II and III, which were evenly matched with the subjective classification.

These discrepancies suggest that subjective classification criteria, particularly among optometrists, may be influenced more by optometric data, clinical experience and visual symptoms than by the mathematical characteristics of the FDC itself. Optometrists rely on their knowledge of fixation disparity and its clinical implications, which may lead them to prioritise empirical patterns over strict mathematical classifications. In contrast, mathematicians, who approach the problem from a purely analytical perspective, assess FDCs solely on their functional properties, leading to greater consistency with the developed model. For cases P_{14} and P_{17} where classification ambiguity remained even within the model, this difference in perspective is evident.

The model developed in this study considered some theoretical background that must be taken into account. First, it considered $-10 \leq a_0, a_2 \leq 10$ for Type I and $-10 \leq a_0, a_3 \leq 10$ for Type IV in order to fix the centre of symmetry between $-10/10$ in the 'x' axis and $-10/10$ in the

'y' axis. This indicates that the model limits the fixation disparity to ≤ 10 arcmin. However, it should be noted that the proposed method can easily accommodate such marginal and uncommon fixation disparities >10 arcmin by simply adjusting values 10 and -10 as appropriate. On the other hand, we established $f_2(x) \geq -10$ for Type II, $f_3(x) \leq 10$ for Type III and $-10 \leq f_4(x) \leq 10$ for Type IV. Note that, in general, this approach ensures a coherent model when the values lie within these ranges. Clinically, this limited range is supported by the literature, which shows that the centre of symmetry for Type I and Type IV typically falls within these limits. Moreover, it is uncommon to observe the asymptotic line exceeding $+10$ arcmin (eso) for Type II or falling below -10 arcmin (exo) for Type III.^{1,3,5,12}

Other elementary functions were also explored to describe each type according to Figure 1. However, after conducting a series of experiments, it was determined that the functions proposed in this method are the most suitable based on the analysis. For instance, the function $f_1(x) = a + bx + cx^2 + dx^3$ was considered initially for Type I, which assumes only one inflection point. Nevertheless, the results obtained with this function were equivalent to those achieved with the proposed function $f_1(x) = a_0 + a_1(x - a_2)^3$.

Another question that must be discussed is how many points are necessary to model a FDC accurately, and does the classification work better with more points? According to both the Wesson fixation disparity card (bernell.com) manual and Scheiman and Wick,³¹ a total of seven points (0, 3BI/3BO, 6BI/6BO and 12BI/12BO) are sufficient for the rapid determination of the shape and centre of symmetry. However, the present study used different vergence demands to determine the curves (0, 5BI/5BO, 10BI/10BO and 15BI/15BO). Although slightly different in terms of magnitude, the number of points remained the same. Indeed, greater discrepancy is found in previous publications, with different points being used to determine the FDC. For example, Hashemi et al. used 0, 3BI/3BO, 6BI/6BO, 9BI/9BO and 12BI/12BO (10 points in total),³² while Frantz and Scharre used 0, 3BI/3BO and 6BI/6BI (total of five points)¹⁹ and Daum et al. used 0, 3BI/3BO, 6BI/6BO and 12BI/12BO (total of seven points).³³ Although it appears that increasing the number of vergence demand values improves the function fit, this statement has limitations with mathematical modelling.

It should be pointed out that Runge's phenomenon could occur when increasing the number of vergence demand values to compute the best-fit polynomial function.³⁴ This phenomenon establishes that oscillations may arise when using high-degree polynomials, which can lead to a decrease in interpolation accuracy rather than an improvement. Note that, in general, a polynomial of degree n is required to obtain a polynomial function that passes through $n + 1$ given points. While it is true that the developed model can incorporate additional points to adjust the elemental functions better and improve the approximation for each type of FDC, using 7–10 points was sufficient

to capture the basic characteristics of the FDC from a mathematical perspective and ensure that the curves remain representative and interpretable in a clinical context.

The slope of a FDC is an important variable for both research and clinical purposes. In this study, slope calculations showed significant differences when using a general third-order polynomial fit compared with those obtained from the derived elemental functions. These findings underscore the importance of fitting FDCs accurately using appropriate elemental functions rather than relying on general-purpose models. Proper classification and function fitting are crucial because they directly influence the resulting slope values and consequently the interpretation of the data. An inaccurate fit may yield incorrect slope values, potentially leading to misleading conclusions in scientific studies or ineffective assessments in clinical settings.

It is important to clarify that, in this context, the 'slope' refers to the variation in fixation disparity between two vergence demand points, typically within the central region of the curve. This is a distinct difference rather than a derivative at a specific point. While the term 'slope' may imply a derivative in mathematical terms, it is used here to represent the average rate of change between two vergence demands.

As noted, function $f_4(x)$ exhibits a steeper slope around the origin due to its intrinsic shape, while $f_1(x)$, $f_2(x)$ and $f_3(x)$ show more gradual changes in this region. This behaviour is illustrated visually in Figures 2 and 4, where the contrast between the slope patterns of the models is apparent.

Although the primary aim of this study was to develop mathematical models for FDCs, these models were derived from a specific sample. As such, although they capture key mathematical characteristics, they are not intended to replace a comprehensive visual assessment, which remains essential for a thorough evaluation of binocular status in clinical practice. Although parameters such as the centre of symmetry or slope can be measured clinically, the proposed mathematical model considers the reliability and objectivity of these measurements by fitting each FDC type with an appropriate elementary function, taking into account the characteristics of the centre of symmetry. This approach reduces bias when classifying the FDC type subjectively and provides an accurate slope calculation. Such objectivity can support clinical decision-making by offering an additional layer of evidence in cases where the FDC type is ambiguous or where clinical signs do not clearly align with patient symptoms.

Specifically, factors such as the presence and severity of visual symptoms, the near point of convergence, heterophoria, fusional vergence reserves and accommodative function play integral roles in the accurate diagnosis of binocular vision anomalies.^{7,12,16} FDCs provide valuable insights into vergence dynamics and disparity-driven adaptation, but they do not capture the full spectrum of oculomotor function. Therefore, while the mathematical model developed here enhances the robust analysis of FDCs, it must be complemented with clinical assessments that incorporate these additional parameters to ensure a better clinical

approach to visual diagnosis and management. Moreover, the experimental FDCs obtained in this study were obtained using a Wesson card; other instruments can be used, such as the Saladin near point balance card or Sheedy disparometer, but these methods are not interchangeable.^{2,16,17} In addition, although other techniques can achieve better precision, the mathematical model for each type of FDC developed in the present study can be extrapolated.

Overall, this study developed and validated a mathematical model to characterise the different types of FDCs, providing an objective and robust tool for both clinical and research applications. The findings highlight the importance of integrating mathematical models to improve the accuracy and consistency of FDC classification. A larger and more diverse dataset would be beneficial for evaluating the generalisability and robustness of the proposed model across different populations and clinical conditions.

AUTHOR CONTRIBUTIONS

Marc Argilés: Conceptualization (lead); data curation (lead); methodology (equal); resources (equal); software (equal); supervision (equal); validation (equal); writing – original draft (equal); writing – review and editing (equal).
Xavier Molinero: Formal analysis (lead); methodology (equal); resources (equal); software (equal); supervision (equal); validation (equal); writing – original draft (equal); writing – review and editing (equal).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

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